Deliverable 3: Sequence Diagram & Detailed Walkthrough

Project: MedSchedule Healthcare Scheduling System | **Date:** August 2026

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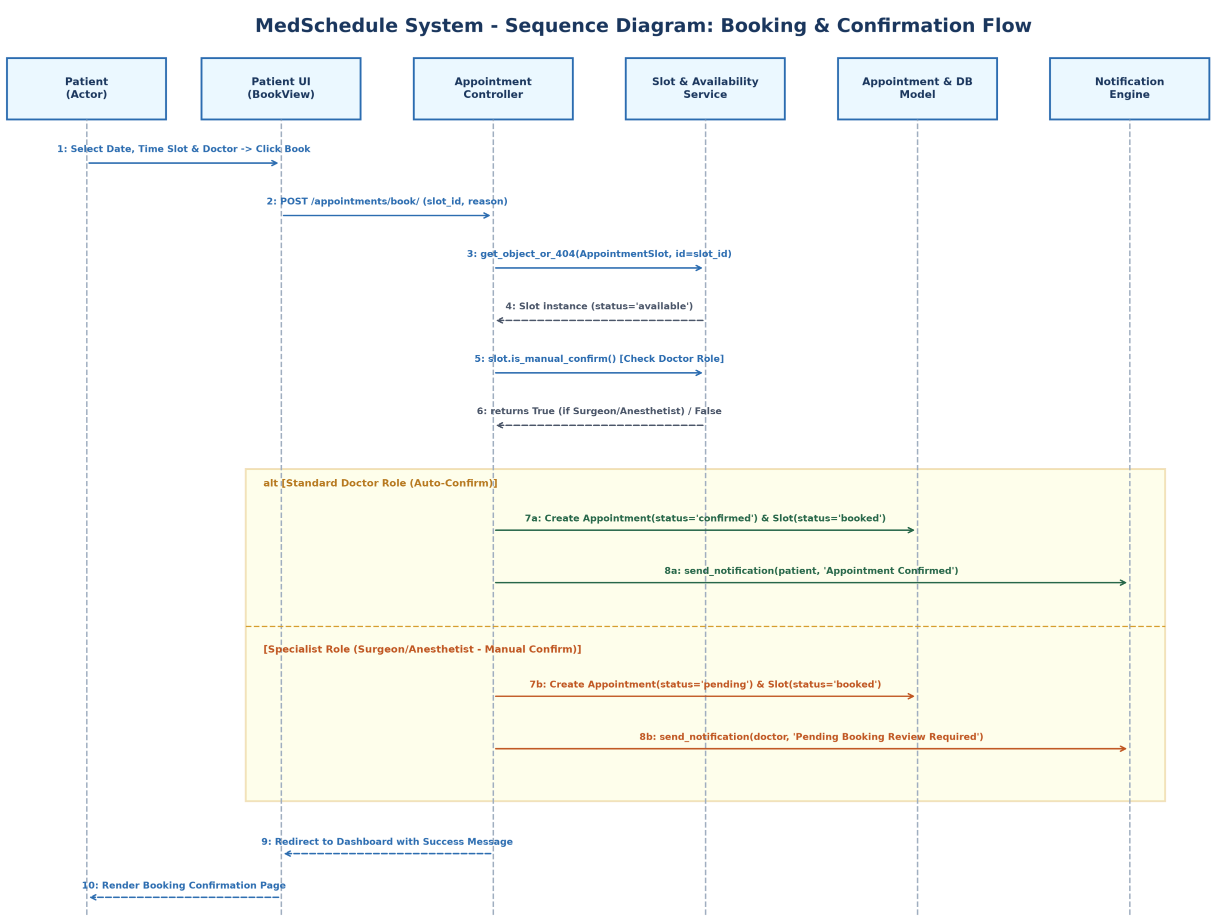
1. Dynamic Behavior & Interaction Scope

The Sequence Diagram models the chronological flow of messages, method invocations, and database state transitions during the critical end-to-end process: **Patient Appointment Booking with Role-Based Auto/Manual Confirmation**.

2. Sequence Diagram: Booking & Confirmation Flow



3. Lifeline Component Definitions

Lifeline Name	Architecture Layer	Role & Implementation Details
Patient (Actor)	Presentation / User	Human user initiating the HTTP request via web interface.
Patient UI (BookView)	Django Template / View	HTML form submitting POST request with slot_id and reason.
Appointment Controller	Django View Controller	schedules.views / appointments.views processing business rules.
Slot & Availability Service	Domain Service / ORM	AppointmentSlot model verifying availability and manual confirm rules.
Appointment & DB Model	Persistence Layer	SQLite Database storing Appointment and updating slot state.
Notification Engine	Utility Service	notifications.utils.send_notification executing SMS/Email dispatches.

4. Step-by-Step Message Execution Sequence

- **Step 1: HTTP Post Submission:** Patient submits slot selection and clinical reason to BookView.
- **Step 2: Controller Dispatch:** BookView validates form data and invokes AppointmentController.book_slot().
- **Step 3: Slot Retrieval & Validation:** Controller queries AppointmentSlot.objects.get(id=slot_id) and verifies status=='available'.
- **Step 4: Role-Based Check:** Controller evaluates slot.is_manual_confirm() by checking if doctor's role is in ['surgeon', 'anesthetist'].
- **Step 5a: Auto-Confirmation Branch (Standard Roles):** If False, status='confirmed'. Appointment object created; slot status updated to 'booked'.
- **Step 5b: Manual Confirmation Branch (Specialist Roles):** If True, status='pending'. Appointment created; slot updated to 'booked' awaiting doctor review.
- **Step 6: Asynchronous Notification Trigger:** send_notification() called with recipient, message text, and delivery preference (email/sms).
- **Step 7: UI Response & Redirection:** Controller returns HTTP 302 redirect to patient dashboard with toast notification.